

Jay Niketan Pathare

jayadmit456@gmail.com | NY, US | Portfolio | LinkedIn | GitHub

EXPERIENCE

Graduate Student Developer | HeinOnline (CSE 611 Project), Buffalo, NY

January 2026 – April 2026

- Engineered a fault-tolerant OCR/LLM metadata extraction pipeline (Tesseract-OCR, Qwen-VL:8b, Qwen-35B) across three independent sources: CrossRef API, web scrapers, and OCR/LLM vision to extract article titles, authors, DOIs, page ranges, and ToC data from law journals (30k+ pages).
- Built a fuzzy-matching engine to merge outputs into a validated structure, cutting manual data-entry work by ~90% and pushing extraction accuracy to ~91% over five iterations, benchmarked against ground truth.
- Parallelized the pipeline with a worker pool that pulls the next journal from a shared queue on completion, and added an “aiscore” system to flag low-confidence fields for human review.

Software Engineer | Thesis Mumbai Tech (Startup), Mumbai, India

August 2024 – August 2025

- Led delivery of platform modules end to end: gathered requirements in client meetings, prototyped to validate scope, and broke the finalized work down across a small team.
- Reviewed and merged team pull requests alongside my own code, catching issues and maintaining code quality.
- Built and scaled healthcare modules (Patient Management, Consent) supporting 10k+ records on normalized relational schemas for fast retrieval and reporting, and designed a real-time WebSocket + Redis pipeline feeding IoT baby-warmer sensor data into PostgreSQL with ~2s live monitoring.
- Delivered a real-time telemedicine consultation feature (WebRTC) connecting on-site nurses with remote doctors to review test results and procedures with patients, alongside secure PDF report generation, and added a color-blindness diagnostic module (95%+ accuracy) with automated reporting for practitioners.
- Containerized 3+ production projects using Docker and Docker Compose, keeping local, staging, and deployment environments consistent and reducing new-developer setup time by ~90%.

Cloud Engineer Intern | Data Maven Pvt Ltd, Mumbai, India

November 2023 – May 2024

- Designed and deployed scalable AWS infrastructure (EC2, RDS) with VPC networking and security groups for secure, low-latency communication between backend services.
- Automated cloud resource provisioning, cutting setup time by ~40% and reducing deployment issues.

Data Engineer Intern | Go Digital Technology Consulting, Mumbai, India

June 2023 – August 2023

- Developed an ETL workflow using Python (Pandas, NumPy) and MySQL to process and clean large business datasets, producing reports on key operational trends.

ACADEMIC PROJECTS

LedgerFlow: Event-Driven Payment Ledger | Java, Spring Boot, Kafka, PostgreSQL

October 2026

- Implemented a payment ledger that posts each payment exactly once even when messages are duplicated or retried, using a transactional outbox, idempotent consumers, and double-entry accounting.
- Prevented overspending under concurrent requests with ordered PostgreSQL row locks: in a 200-request race on an account funded for 50 payments, exactly 50 succeeded, 150 were safely rejected, and the balance ended at \$0.
- Verified with 94 automated tests and a 27/27-run chaos campaign (18,000 duplicate events, 14 real JVM crashes, 0 invariant violations); a local Kubernetes (k3d) pod-restart demo completed 250/250 payments with 0 duplicate postings.

RIPPLE: Agentic Change-Impact Analysis | Python, Pydantic, Git, LLM API

September 2026

- Developed RIPPLE, an AI agent (Python CLI) that predicts which files, tests, and regression risks a feature request will touch before any code is written, using 7 static-analysis tools, a 25-tool-call cap, and Python-verified evidence.
- Evaluated on 30 unseen software-change tasks from 16 repositories (90 runs), reaching 0.306 precision and 0.311 MRR, with 0 unsupported claims accepted into any report.

LLM Application Optimization Engine | Python, Qwen (Novita API), Groq API, pytest, LaTeX

August 2026

- Engineered a multi-model pipeline that turns a job description into a tailored one-page resume, with Python checking every generated claim against source evidence and rejecting unsupported ones, backed by a 2,500+ test pytest suite.
- Cut typical end-to-end runtime by ~73% (about 11 minutes to about 3) using three concurrent generation workers, regeneration of only failed parts, and local fixes in place of repeat model calls.

EDUCATION

Master's in Computer Science

August 2025 – February 2027

University at Buffalo, The State University of New York, Buffalo, NY

GPA: 3.85 / 4.0

Bachelor of Engineering, Information Technology

Vivekanand Education Society's Institute of Technology, Mumbai University, India

August 2019 – May 2023

TECHNICAL SKILLS

- Languages & Databases:** Python, Java, JavaScript, Go, SQL, PostgreSQL, MySQL
- Frameworks:** Django, Django REST Framework, Spring Boot, ReactJS, Pandas, NumPy, PyTorch, scikit-learn, Pydantic
- AI/ML:** LLM Integration, Prompt Engineering, Machine Learning, Deep Learning, Agentic AI, AI Agents, Tool Calling, Model Evaluation, Neural Networks (CNN, GNN), Transformers
- Software Engineering:** OOP, REST APIs, Full-Stack Development, Backend Development, System Design, ETL, Parallel Processing, Agile, Real-Time Monitoring, Event-Driven Architecture, Distributed Systems, Apache Kafka
- Tools:** Linux, Redis, AWS (EC2/RDS), Docker, Git, GitHub, GitHub Copilot, WebSockets, OAuth/Auth0, Kubernetes

CERTIFICATIONS, AWARDS & PUBLICATIONS

- 1st Place, ACV Auctions + Copart Challenge** | AI for Good Hackathon Design Challenge, UB October 2026
- AWS Certified Cloud Practitioner** July 2024
- Research Publication:** J. Pathare et al., “Music Genre Classification,” IJRAR, Apr. 2023 (97.68% accuracy). [Link](#).